Assignment Quiz

Course: B.Tech | Stream: Computer Science | Subject: Data Structure & Algorithm | Sem: 3rd | Paper Code: PCC-CS301

Unit 5

Topics: Comprehensive: Introduction & Searching; Stacks & Queues; Linked Lists & Trees; Sorting/Hashing/Graphs | 10 MCQ Questions | Time: 5 minutes

Q1. Which notation describes the running time most precisely when the best and worst cases coincide?

a) Big-O
b) Omega
c) Theta
d) None of these

Correct Answer: c

Q2. Which data structure is used to implement Breadth-First Search (BFS)?

a) Stack
b) Queue
c) Tree
d) Hash Table

Correct Answer: b

Q3. What is the time complexity of accessing an element by index in an array?

a) O(1)
b) O(n)
c) O(log n)
d) $O(n^2)$

Correct Answer: a

Q4. Which sorting algorithm is best suited when memory space is limited and stability is not required?

a) Merge Sort
b) Quick Sort
c) Counting Sort
d) Bucket Sort

Correct Answer: b

Q5. A hash table with a good hash function and low load factor provides average-case search time of:

a) O(n)
b) O(log n)
c) O(1)
d) $O(n^2)$

Correct Answer: c

Q6. Which traversal of a tree is used to delete the entire tree safely (children before parent)?

a) Pre-order
b) In-order
c) Post-order
d) Level-order

Correct Answer: c

Q7. What is the worst-case time complexity of Linear Search on an array of n elements?

a) O(1)
b) O(log n)
c) O(n)
d) O(n^2)

Correct Answer: c

Q8. Which data structure allows both insertion and deletion from both ends?

a) Stack
b) Simple Queue
c) Deque (Double-ended Queue)
d) Singly Linked List

Correct Answer: c

Q9. In an AVL Tree, a rotation is triggered when the balance factor of a node becomes:

a) 0
b) +1 or -1
c) +2 or -2
d) It never triggers automatically

Correct Answer: c

Q10. Which of the following best describes a spanning tree of a graph?

a) A subgraph that includes all vertices, is connected, and contains no cycles
b) A tree that stores only leaf nodes
c) A binary search tree derived from graph edges
d) A graph with no vertices

Correct Answer: a

--- End of Quiz ---